COMICCRAFT USING AI

SAHAYA RUBERT E

1. Introduction

ComicCraft Using AI is a web-based application that helps users create personalized comic stories using Artificial Intelligence. The main purpose of this project is to convert a simple story idea into a complete comic with multiple panels, narration, character dialogues, and illustrations.

The user provides important details such as the story prompt, main character name, setting, story tone, and art style. The application processes these details using AI models and generates a panel-by-panel comic story. The final comic can be viewed on a web page and downloaded as a PDF.

ComicCraft is designed to make comic creation easier for people who may not have drawing or professional storytelling skills. It combines AI-based text generation with AI image generation to create a complete comic from a simple idea.

2. Objectives of the Project

The main objectives of ComicCraft Using AI are:

- To create personalized comic stories using AI.
- To generate a structured five-panel comic outline.
- To create narration and character dialogues automatically.
- To generate illustrations for each comic panel.
- To provide a simple and user-friendly web interface.
- To display the generated comic panel by panel.
- To export the completed comic as a PDF.
- To provide both a web interface and API endpoints for comic generation.

3. Technologies Used

The project uses several technologies to complete the comic generation process.

FastAPI is used as the backend framework. It manages the application routes, receives user inputs, communicates with the AI services, and sends the generated results to the frontend.

HTML, CSS, and Jinja2 are used to create the frontend. Jinja2 allows the application to display the AI-generated comic information dynamically.

Google Gemini AI is used for story generation. Gemini can generate structured comic outlines, narration, scene descriptions, and character dialogues based on the user's input.

Hugging Face image generation is used for creating illustrations from the image prompts generated for each panel.

FPDF is used to create the final PDF. It combines the comic images and story content into a downloadable document.

Python is used as the main programming language for the backend and AI integration.

4. Working of the Project

The ComicCraft workflow starts when the user opens the web application. The home page contains a form where the user enters the required information.

The user provides a story prompt, character name, setting, tone, and art style. For example, the user can enter a story about a brave fox exploring an enchanted forest and select a funny tone and comic-book art style.

After submitting the form, the FastAPI backend receives the information. The application first creates a five-panel outline. Each panel contains information such as the panel number, title, scene description, and image-generation prompt.

Next, the AI generates detailed narration and character dialogues for the panels. This makes the comic story more complete and engaging.

After the text content is created, image-generation services create an illustration for each panel. The generated images are connected with the corresponding panel information.

The application then builds the final comic layout. Each panel contains its title, image, scene description, captions, narration, and other required information.

Finally, the complete comic can be exported as a PDF. The PDF contains the generated images and story content and can be downloaded by the user.

5. Main Features

ComicCraft provides several important features.

The **Comic Generation Form** allows users to enter their creative ideas and customize the comic.

The **Five-Panel Story Generation** feature creates a structured comic story with five panels.

The **AI Story Generation** feature creates narration and character dialogues automatically.

The **AI Image Generation** feature creates illustrations based on the scene and art-style requirements.

The **Comic Preview** page displays the generated panels in sequence so that the user can read and view the complete story.

The **PDF Export** feature allows users to download the completed comic for saving, printing, or sharing.

The application also provides API endpoints that can be used to test and access comic-generation functions directly.

6. Advantages

ComicCraft reduces the time required to create a comic manually. A user only needs to provide a simple idea and some preferences. The AI system can then generate the story structure, narration, dialogue, and images.

The application is also useful for beginners because users do not need advanced drawing or professional comic-writing skills. The simple web interface makes the process easier to understand.

Another advantage is customization. Users can change the character, setting, tone, and art style to create different types of comics.

7. Conclusion

ComicCraft Using AI demonstrates how Artificial Intelligence can be used for creative storytelling. The project combines a FastAPI backend, web technologies, Gemini AI, image generation, and PDF creation into one application.

The system provides an end-to-end process from a simple user idea to a complete illustrated comic. It generates a structured story, creates narration and dialogues, produces illustrations, displays the comic through a web interface, and allows the final result to be exported as a PDF.

This project shows the practical use of Generative AI in creative applications. In the future, the project can be expanded with user accounts, comic libraries, saved projects, character customization, and longer multi-page stories.